

HU Collection

TMUA Mock Set F Paper 2

Worked Solutions

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Answer key

Q	1	2	3	4	5	6	7	8	9	10
Ans	C	C	A	E	D	D	B	A	F	B
Q	11	12	13	14	15	16	17	18	19	20
Ans	B	C	F	D	B	E	A	D	E	E

Each solution gives the intended route and names the trap the question was built around. Questions are ordered by increasing difficulty.

Question 1. Answer C.

Translate each clause. “ A only if B ” is $A \Rightarrow B$. “ C if B ” is $B \Rightarrow C$. “Not C unless A ” means “if not A then not C ”, whose contrapositive is $C \Rightarrow A$. The implications form a cycle $A \Rightarrow B \Rightarrow C \Rightarrow A$, so all three are equivalent.

Question 2. Answer C.

Last digits repeat with period 4 (or 2 for 4). Powers of 2 end in 2, 4, 8, 6, and $2024 \equiv 0 \pmod{4}$ gives 6: true. Powers of 3 end in 3, 9, 7, 1, and $2026 \equiv 2$ gives 9: true. Powers of 7 end in 7, 9, 3, 1, and $2025 \equiv 1$ gives 7: true. Powers of 4 end in 4, 6 alternately, and 2025 is odd, giving 4: false.

Question 3. Answer A.

I is correct. II divides $x^2 + 1 \geq -2x$ by the negative number x and correctly reverses it: $x + \frac{1}{x} \leq -2$. III combines the two cases. The proof is valid. Using $(x - 1)^2$ in case II would only give $x + \frac{1}{x} \leq 2$, which is true but too weak; choosing $(x + 1)^2$ is what makes it work.

Question 4. Answer E.

S is true, since $4 \mid n$ gives $16 \mid n^2$, and so is its contrapositive. The converse, “if $8 \mid n^2$ then $4 \mid n$ ”, is also true: $8 \mid n^2$ forces n even, and writing $n = 2m$, $8 \mid 4m^2$ forces m even. The inverse is the contrapositive of the converse, so it is true as well. All four hold. The instinct that “the converse is usually false” is exactly what this question tests.

Question 5. Answer D.

$x^3 - x = x(x - 1)(x + 1) < 0$ exactly when $x < -1$ or $0 < x < 1$. So I and IV are sufficient. II and III both allow $-1 < x < 0$, where for instance $x = -\frac{1}{2}$ gives $x^3 = -\frac{1}{8} > -\frac{1}{2}$.

Question 6. Answer D.

I: $\frac{k^2 ab}{k(a+b)} = k \cdot \frac{ab}{a+b}$ is an integer. II: $2a \mid a^2$ exactly when $2 \mid a$. III: $b + 1 \mid b$ is impossible for positive b , since $b + 1 > b$. IV fails: $(10, 15)$ has $25 \mid 150$, but neither number divides the other.

Question 7. Answer B.

“Hence N is prime” does not follow: $2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13 + 1 = 30031 = 59 \cdot 509$. What is true is that $N > 1$ has some prime factor, and no p_i divides N , so that factor is a prime missing from the list. That still contradicts the assumption, so the proof survives once that one step is corrected.

Question 8. Answer A.

I: a prime dividing ab divides a or b , and if it also divides $a + b$ it divides both, which is impossible. II: a common divisor of $a + b$ and $a - b$ divides their sum $2a$ and difference $2b$, hence divides $2\gcd(a, b) = 2$. III: a^2 and b^2 have the same prime factors as a and b . IV fails for $a = b = 1$, since $\gcd(2, 2) = 2$; in general $a^2 + b^2 = (a + b)^2 - 2ab$, so the common divisor divides $2ab$ and can be 2.

Question 9. Answer F.

I: if $a + b$ were rational, then $b = (a + b) - a$ would be rational. II fails for $a = 0$. III fails for $x = \sqrt{2}$, $y = -\sqrt{2}$. IV is the contrapositive of “if x is rational then x^2 is rational”, which is true. So I and IV.

Question 10. Answer B.

A and B are of opposite types, since A’s statement is true exactly when B is a knave. Suppose B is a knight. Then A is a knave, and exactly one knight (B) means C is a knave; but C’s claim that B and C differ would then be true, which a knave cannot say. So B is a knave and A is a knight. B’s statement is false, so the number of knights is not one; with A a knight, C must be a knight too. Check C: B and C do differ, so C tells the truth.

Question 11. Answer B.

I: squaring shows $\sqrt{x+y} \leq \sqrt{x} + \sqrt{y} \iff x+y \leq x+y+2\sqrt{xy}$, which is true. III: $x+y+1 \leq x+y+2$. V gives equality, which is allowed. II fails at $x = y = 1$ ($4 > 2$), and IV fails at $x = y = 2$ ($16 > 8$). So I, III and V.

Question 12. Answer C.

I: the third angle is also 60° . II only forces the middle angle to be 60° (for example $30^\circ, 60^\circ, 90^\circ$). III: take sides $a - d, a, a + d$. If the 60° angle faces a , the cosine rule gives $a^2 = a^2 + 3d^2$; if it faces $a + d$ or $a - d$, it gives $3ad = 0$. Either way $d = 0$. IV reduces to III, since angles in AP include a 60° angle.

Question 13. Answer F.

A continuous non-negative function with zero integral over $[0, 2]$ must vanish throughout $[0, 2]$: if it were positive somewhere, it would be positive on a small interval and the integral would be positive. So I holds. A polynomial with infinitely many roots is the zero polynomial, so II holds, and then III. IV fails for a continuous function that is zero on $[0, 2]$ and positive beyond it, such as $\max(0, x - 2)$, which breaks II and III.

Question 14. Answer D.

Squares that are also cubes are sixth powers: 1 and 64. 64 fails (4) and (5), and 1 fails (3) and (5), so neither works with only one false statement. So exactly one of (1) and (2) is false and the other

four are true. If (1) is false, n is an even cube below 50 divisible by 9: none, since $8 \nmid$ and 27 is odd. If (2) is false, n is an even square below 50 divisible by 9: that is 36.

Question 15. Answer B.

f' is a quadratic that changes from positive to negative at $x = 2$, and $f'(0) > 0$. If f' opens upwards, 2 is its smaller root and the minimum is to the right; if it opens downwards, 2 is its larger root and the other root is below 0. Either way $f' > 0$ on $[0, 2)$, so III holds and $f(2) > f(0) = 0$ gives II. I fails in the second case. IV can fail too: $f(x) = x^3 - 9x^2 + 24x$ has $f'(x) = 3(x - 2)(x - 4)$, so $f'(0) = 24 > 0$ and a maximum at 2, but $f(x) = x(x^2 - 9x + 24)$ has only the real root 0.

Question 16. Answer E.

II: on $(0, \frac{\pi}{2})$, $\tan x > x$. On $(\frac{\pi}{2}, \pi]$, $\tan x \leq 0 < x$, so any solution lies in $(\pi, \frac{3\pi}{2})$, where $\tan x - x$ increases from $-\pi$ to $+\infty$: exactly one solution, and it exceeds π . So I and III hold. IV fails: at $x = \frac{5\pi}{4}$, $\tan x = 1 < \frac{5\pi}{4}$, so the crossing comes later (in fact near 4.49).

Question 17. Answer A.

Compare each with 1.5 and then refine. $2^{1.5} = 2\sqrt{2} \approx 2.83 < 3$, so $p > 1.5$. $3^{1.5} = 3\sqrt{3} \approx 5.20 > 5$, so $q < 1.5$. To separate q and r : $5^{1.3} > 8$ (since $5^{13} \approx 1.22 \times 10^9 > 8^{10} \approx 1.07 \times 10^9$), so $r < 1.3$, while $3^{1.4} < 5$ (since $3^{14} = 4\,782\,969 < 5^{10} = 9\,765\,625$), so $q > 1.4$. Hence $r < q < p$.

Question 18. Answer D.

Oddness gives $f(0) = 0$, and then $f(4) = f(4 - 0) = f(0) = 0$: I. Next $f(x + 4) = f(4 - (x + 4)) = f(-x) = -f(x)$, and applying this twice gives $f(x + 8) = f(x)$: II. IV: $f(6) = f(4 - 6) = f(-2) = -f(2)$. III fails: $f(x) = \sin \frac{\pi x}{4}$ satisfies both conditions and has $f(2) = 1$.

Question 19. Answer E.

P holds when $k^2 \geq 4$, that is $k \leq -2$ or $k \geq 2$. Q holds when $1 - 4k \geq 0$, that is $k \leq \frac{1}{4}$, so $k \leq 0$ for integers. Exactly one holds when P is true and Q false, which is $k = 2, \dots, 10$ (nine values), or Q is true and P false, which is $k = -1, 0$ (two values). That is 11. For $k \leq -2$ both hold, and for $k = 1$ neither does.

Question 20. Answer E.

I: $0 = (a + b + c)^2 = 6 + 2(ab + bc + ca)$. II: substituting $c = -a - b$ gives $b^2 + ab + a^2 - 3 = 0$, and for real b the discriminant $a^2 - 4(a^2 - 3) = 12 - 3a^2 \geq 0$ forces $|a| \leq 2$, and likewise for b and c . IV: $\sum a^2 b^2 = (ab + bc + ca)^2 - 2abc(a + b + c) = 9$, so $\sum a^4 = 36 - 2 \cdot 9 = 18$. III fails: $a = 2$, $b = c = -1$ satisfies both conditions and has $abc = 2$.